Closed-Loop Calibrated Robustness for Tube MPC: Risk-Budgeting under Feedback Covariate Shift

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***Abstract*—Controllers built on learned models face a critical safety question: once deployed, how often will they actually violate state or input constraints? While existing conformal-control methods successfully certify prediction-set coverage or planner-level feasibility, they rarely quantify closed-loop, mission-level safety. This paper bridges this gap for linear tube model predictive control (MPC) subject to additive model errors.**

We introduce a certified *risk map* that, for any operating context and candidate tube radius, bounds the probability that the next model error escapes the tube. An online budget manager then spends a user-specified mission risk budget through predictable local certificates. We prove that finite tables of local miss probabilities can be calibrated with finite-sample guarantees using supplied likelihood ratios or certified bounded upper weights for the conditional residual laws queried under feedback. On the calibration event, hard-feasible implemented plans whose spent certificates sum to at most ε yield a first-step violation count with conditional expectation at most ε, with a companion martingale tail bound. We also bound the MPC performance loss caused by using certified rather than oracle miss curves. A tube-MPC benchmark examines risk attainability and the cost of certification under feedback-dependent context visitation and prescribed within-context reweighting.

***Index Terms*—Robust model predictive control, learning-enabled control, conformal prediction, feedback covariate shift, uncertainty quantification, finite-sample guarantees.**

I. Introduction

SAFETY guarantees for learning-enabled control are often stated at a partially satisfying level. A prediction method may certify that a future model error lies in a set with high probability, or a planning method may certify that its optimization problem remains feasible. Both are useful, but a control engineer usually wants a more direct statement: *over the whole mission, how often can the deployed controller violate safety constraints?*

This question becomes difficult when the controller changes the state–input distribution seen at deployment. We call this *feedback-induced covariate shift*: model-error samples used for calibration are collected under one closed-loop policy or operating mix, whereas the deployed feedback policy sequentially drives the future covariates and hence may query different model-error laws [1], [2]. Our analysis takes valid conditional-law envelopes as statistical inputs. We use a *context* $g \in \mathcal{G}$ as a finite label grouping states, inputs, or prediction stages with similar model-error statistics; we refer to the corresponding group as an *operating region*. A *miss* is the event that the next model error leaves the disturbance set selected for that context and radius; a local certificate is *predictable* if it is chosen from information available before that error is realized. Thus the mission-level bound discussed in the sequel concerns the accumulated realized first-step state/input violations of the receding-horizon loop, not merely open-loop prediction-set coverage. Related challenges also arise in data-driven and learning-based control, where controllers must remain safe despite uncertainty in the disturbance distribution; see, e.g., [3], [4]. This paper addresses the gap for linear or locally linear tube MPC with additive model error and context-dependent ellipsoidal disturbance sets. The key object is a *risk map*: for each operating region and candidate tube radius, it upper-bounds the probability of a miss under the deployment law.

The paper makes three contributions. First, it constructs finite-sample risk maps under a supplied bounded envelope for deployment model-error law shift. Second, it shows that predictable local miss-probability certificates add up to a mission-level bound on realized first-step violations. Finally, it quantifies the performance loss induced by using certified rather than oracle miss curves.

A. Relation to Existing Conformal-Control Methods

Weighted conformal MPC uses likelihood-ratio weighting to construct fixed-coverage uncertainty sets or quantiles inside MPC [5], [6]; shrinking-horizon, feedback-conformal prediction (CP), and system-level-synthesis (SLS)-MPC variants certify recursive feasibility, mission-risk reallocation, or state-dependent ellipsoids [7]–[9]. Here the calibrated object is different: for each context–radius pair, it upper-bounds the deployment miss probability $r_g(\lambda)$, see Definition 1 below, producing a finite miss-probability map that the controller spends online.

The main novelty is the combination of weighted local risk maps, tube-MPC radius selection, and mission-level risk bookkeeping. Other shift-aware control/planning methods use out-of-distribution (OOD)-triggered fallbacks, sequence-model conformal risk control, region-wise robust bounds, or adversarially robust transfer across policy updates [10]–[14]. These approaches are complementary to ours; the present scope is additive-error tube MPC once contextwise miss certificates are available. Statistically, the finite catalog is related to finite-class or group-conditional conformal guarantees [15]: contexts are pre-specified bins and radii form a finite decision class. We certify binary tube-escape risks and use them in a control budget. Conformal risk control and decision-aware variants [16]–[18] are closest in spirit to the downstream radius-selection view.

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II. Problem Formulation and Budgeted Controller

A. Tube MPC and contexts

Throughout, $t = 0{:}T-1$ indexes a mission of length T and $\ell = 0{:}H-1$ the MPC prediction horizon. The filtration $\mathcal{F}_t$ contains offline design/calibration data and known references, together with the observed closed-loop history through x_t. Choices $g_t, \alpha_t, \lambda_t, p_t$ are $\mathcal{F}_t$-measurable, before the next error is realized. Consider a linear or locally linear prediction model used by the controller for planning

$$x_{t+1} = A_t x_t + B_t u_t + e_t, \tag{1}$$

where $x_t \in \mathbb{R}^n$, $u_t \in \mathbb{R}^m$, and $e_t \in \mathbb{R}^n$ is the additive model error. State and input constraints are polytopic,

$$\mathcal{X} = \{x : H_x x \le h_x\}, \qquad \mathcal{U} = \{u : H_u u \le h_u\}.$$

At time t, tube MPC computes a nominal trajectory

$$z_{t,0} = x_t, \qquad z_{t,\ell+1} = A_{t,\ell} z_{t,\ell} + B_{t,\ell} v_{t,\ell},$$

with $\tilde{x}_{t,0} = x_t$ and candidate tube policy $\tilde{u}_{t,\ell} = v_{t,\ell} + K_{t,\ell}(\tilde{x}_{t,\ell} - z_{t,\ell})$; only $u_t = v_{t,0}$ is implemented before replanning. For this candidate continuation, define the planning-time residual $e_\ell^{[t]} := \tilde{x}_{t,\ell+1} - A_{t,\ell}\tilde{x}_{t,\ell} - B_{t,\ell}\tilde{u}_{t,\ell}$. Then $\eta_{t,\ell} := \tilde{x}_{t,\ell} - z_{t,\ell}$ obeys

$$\eta_{t,\ell+1} = A_{t,\ell}^K \eta_{t,\ell} + e_\ell^{[t]}, \qquad A_{t,\ell}^K := A_{t,\ell} + B_{t,\ell} K_{t,\ell}.$$

Here $A_{t,\ell}, B_{t,\ell}, K_{t,\ell}$ are fixed planning-time matrices. At $\ell = 0$, they agree with the current model and $e_0^{[t]} = e_t$; later residuals are relative to this frozen plan. The probabilistic guarantee below concerns implemented first moves.

The disturbance sets used by the controller depend on an operating context g and a radius λ. A context is a finite label attached to the current operating regime; it can encode, for example, proximity to active constraints, a region of the nominal state space, or the prediction stage. The only structural requirement used below is that the context catalog is finite: $\mathcal{G} = \{g_1, \dots, g_M\}$. The catalog and labeling rule are fixed before calibration. Let N_g denote the number of offline model-error samples assigned to context g for calibration, and let $\mathcal{L} = \{\lambda_1, \dots, \lambda_L\}$, with $\lambda_1 < \cdots < \lambda_L$, be the finite radius grid. Finer partitions improve local resolution but reduce N_g and increase calibration error; finer grids reduce discretization error at the price of a larger $\log(ML)$ confidence penalty. Any data-driven choice uses an independent design split.

For each $g \in \mathcal{G}$ and $\lambda \ge 0$, define the disturbance set

$$\mathcal{E}(g, \lambda) := \left\{ e \in \mathbb{R}^n : \ \|M_g e\|_2 \le \lambda \right\}, \tag{2}$$

where M_g is nonsingular. Its support function is $h_{\mathcal{E}(g,\lambda)}(c) = \lambda \|M_g^{-\top} c\|_2$. The corresponding tube recursion is standard in tube-based robust MPC [19], [20]:

$$\mathcal{T}_{t,0} = \{0\}, \qquad \mathcal{T}_{t,\ell+1} = A_{t,\ell}^K \mathcal{T}_{t,\ell} \oplus \mathcal{E}(g_{t,\ell}, \lambda_{t,\ell}). \tag{3}$$

At time t, fix the measured-state context g_t. For every candidate radius λ, use the horizon completion $g_{t,\ell} = g_t$ and $\lambda_{t,\ell} = \lambda$ for all $\ell = 0{:}H-1$. Planning matrices and references are held fixed across candidates; all quantities are recomputed at the next step. The inner problem is

$$\begin{aligned} J_t := \min_{\{z_\ell, v_\ell\}} & \sum_{\ell=0}^{H-1} c_{t,\ell}(z_\ell, v_\ell) + c_{t,H}(z_H) \\ \text{s.t. } & z_0 = x_t, \\ & z_{\ell+1} = A_{t,\ell} z_\ell + B_{t,\ell} v_\ell, \quad \ell = 0{:}H-1, \\ & z_\ell \in \mathcal{X} \ominus \mathcal{T}_{t,\ell}, \quad v_\ell \in \mathcal{U} \ominus K_{t,\ell}\mathcal{T}_{t,\ell}, \\ & z_H \in \mathcal{X} \ominus \mathcal{T}_{t,H}, \end{aligned} \tag{4}$$

where $\ominus$ denotes Pontryagin difference. Only hard-feasible plans receive the local miss certificate. Here, a *fallback* is a backup control action used when no certified hard-feasible tube-MPC step is available. Such a step remains covered by the mission guarantee only if the backup policy has its own predictable upper bound $q_t \le B_t$ on the conditional violation probability; the budget update is $B_{t+1} = B_t - q_t$. Without such a certificate, the universal bound $q_t = 1$ allows for one possible violation at that step.

B. Practical objective and first-step safety metric

Because receding-horizon MPC implements only the first move before re-optimizing, the safety quantity of interest is the *first-step* violation indicator

$$V_t := \mathbf{1}\{x_{t+1} \notin \mathcal{X} \text{ or } u_t \notin \mathcal{U}\}. \tag{5}$$

The mission-level count is $N_T := \sum_{t=0}^{T-1} V_t$, including any fallback steps. The calibration data are

$$\mathcal{D} = \{E_{g,i} : g \in \mathcal{G}, \ i = 1{:}N_g\}, \tag{6}$$

where $E_{g,i}$ is the ith offline realization, assigned to context g, of the one-step model error in (1). The tube radius is the scalar λ_t in $\mathcal{E}(g_t, \lambda_t)$. The ideal design problem is to choose a predictable radius policy π solving

$$\min_\pi \ \mathbb{E}_\pi \sum_{t=0}^{T-1} J_t(\lambda_t^\pi) \quad \text{s.t.} \quad \mathbb{E}_\pi[N_T] \le \varepsilon,$$

and the MPC loss relative to an oracle radius $\lambda_t^\star$ is

$$\mathcal{L}_T(\pi) := \mathbb{E}_\pi \sum_{t=0}^{T-1} \big(J_t(\lambda_t^\pi) - J_t(\lambda_t^\star) \big).$$

For the first move, $z_{t,0} = x_t$, hence $u_t = v_{t,0}$; once the nominal plan is feasible, first-step input admissibility is deterministic and the only stochastic source of a violation is the one-step model error e_t. The next implication is the standard first-step tube-MPC containment argument; see, e.g., [19, Sec. 3] and [20, Ch. 3]. We state it explicitly because it is the bridge between robust planning and the violation count (5). The following proposition is proved in Appendix A.

Proposition 1 (Standard first-step robust implication)**.** *Suppose the controller applies the first input of a hard feasible tube-MPC plan based on* (2)–(3)*. Then*

$$V_t \le \mathbf{1}\{e_t \notin \mathcal{E}(g_t, \lambda_t)\} \qquad \textit{a.s.} \tag{7}$$

for the first-step context–radius pair chosen by the plan.

Definition 1 (Certified miss curve)**.** *For each context $g \in \mathcal{G}$ and grid radius $\lambda \in \mathcal{L}$, define the* deployment miss probability

$$r_g(\lambda) := Q_g^\star\{\|M_g E\|_2 > \lambda\},$$

where $Q_g^\star$ is made precise in Section III. For the closed-loop theorem, r_g is equivalently any deterministic envelope satisfying $\mathbb{P}\{e_t \notin \mathcal{E}(g,\lambda) \mid \mathcal{F}_t, g_t = g\} \le r_g(\lambda)$ for all queried histories. This is the probability that the one-step model error falls outside the selected disturbance set. Section III constructs a finite-grid upper confidence bound $\bar{r}_g(\lambda)$ such that, with high probability, $r_g(\lambda) \le \bar{r}_g(\lambda)$ for all $\lambda \in \mathcal{L}$. Non-grid radii are either disallowed or assigned the certified bound of the nearest grid point $\lambda_j \le \lambda$, since miss curves are nonincreasing.

C. Budgeted controller and the local target

The online controller uses a certified risk map together with a remaining mission budget. Starting from $B_0 = \varepsilon$, iterations at each time t are:

1. identify the current context g_t;
2. choose a *local target* $\alpha_t \in [0, B_t]$;
3. query $\lambda \mapsto \bar{r}_{g_t}(\lambda)$ on $\mathcal{L}$ and pick λ_t satisfying $p_t := \bar{r}_{g_t}(\lambda_t) \le \alpha_t$;
4. solve tube MPC with first-step disturbance set $\mathcal{E}(g_t, \lambda_t)$ and apply the first move;
5. update the remaining budget as $B_{t+1} = B_t - p_t$.

Here, α_t is the maximum certified miss probability that the budget manager is willing to allocate at time t, based on the remaining budget B_t. The quantity $p_t := \bar{r}_{g_t}(\lambda_t)$ is the certified miss probability actually spent at time t, namely, the certified upper bound on the probability that the one-step error leaves the disturbance set used by the controller. By construction, $p_t \le \alpha_t$.

Let $\Lambda_t^{\mathrm{mpc}} \subseteq \mathcal{L}$ be the radii for which (4) is feasible at time t. A radius meeting the preselected target exists exactly when

$$\underline{p}_t := \min_{\lambda \in \Lambda_t^{\mathrm{mpc}}} \bar{r}_{g_t}(\lambda) \le \alpha_t. \tag{8}$$

Set $\min \varnothing = +\infty$. If (8) fails, a larger target may be used only within B_t. Otherwise certification stops; subsequent recovery moves are tracked separately as in Section IV. Affordability is checked before applying the exact ledger update. A default target is $\alpha_t = B_t/(T-t)$, but any predictable policy is allowed, and the actual spend is p_t.

The architecture is shown in Fig. 1. Sections III–V supply its three ingredients: how to calibrate $\bar{r}_g$, how the bookkeeping rule controls N_T, and how much performance is lost by using certified rather than oracle miss curves.

III. Risk-Map Calibration under Feedback Covariate Shift

For any recorded transition, define the residual

$$E_\tau := x_{\tau+1} - A_\tau x_\tau - B_\tau u_\tau,$$

with online counterpart $e_t = x_{t+1} - A_t x_t - B_t u_t$. Let Q_g^0 be the calibration law of E_τ conditioned on the offline data-gathering policy visiting context g. The deployment object $Q_g^\star$ is conditional: for every history with $g_t = g$, the law $\mathcal{L}(e_t \mid \mathcal{F}_t, g_t = g)$ equals $Q_g^\star$ or is dominated by the same grid miss probabilities; a merely marginal within-context law is insufficient for Theorem 2. The catalog, labeling rule, radius grid, and matrices M_g are frozen before this calibration split; if selected from data, they require an independent design split or a valid post-selection concentration argument. The data in (6) are assumed i.i.d. within each context for the finite-sample theorem: $E_{g,i} \overset{\text{i.i.d.}}{\sim} Q_g^0$, $i = 1{:}N_g$. Independent calibration experiments satisfy this assumption. The benchmark uses independent design and calibration samples.

For candidate radius λ, the true deployment miss probability is $r_g(\lambda) := Q_g^\star\big(\|M_g E\|_2 > \lambda\big)$. When $Q_g^\star \ne Q_g^0$, the miss rate must be corrected for shift. We do so with likelihood-ratio weights

$$\rho_g(e) := \frac{dQ_g^\star}{dQ_g^0}(e), \qquad 0 \le \rho_g(e) \le \bar{\rho}. \tag{9}$$

The weighted empirical miss rate is

$$\tilde{r}_g(\lambda) := \frac{1}{N_g} \sum_{i=1}^{N_g} \rho_g(E_{g,i}) \mathbf{1}\{\|M_g E_{g,i}\|_2 > \lambda\}.$$

The proof of the following theorem is given in Appendix B.

Theorem 1 (Uniform finite-sample risk map)**.** *Fix a finite context catalog $\mathcal{G} = \{g_1, \dots, g_M\}$, a finite radius grid $\mathcal{L} = \{\lambda_1, \dots, \lambda_L\}$, and matrices M_g and weights fixed independently of the calibration residuals. Assume that for each $g \in \mathcal{G}$, $N_g \ge 2$ and the calibration residuals $\{E_{g,i}\}_{i=1}^{N_g}$ are i.i.d. under Q_g^0 and satisfy* (9)*.*

Define the weighted miss indicators

$$Z_{g,i}(\lambda) := \rho_g(E_{g,i}) \mathbf{1}\{\|M_g E_{g,i}\|_2 > \lambda\} \tag{10}$$

and the associated sample variance

$$s_g^2(\lambda) := \frac{1}{N_g - 1} \sum_{i=1}^{N_g} \Big(Z_{g,i}(\lambda) - \tilde{r}_g(\lambda) \Big)^2.$$

For $\delta \in (0,1)$, define the certified risk map

$$\bar{r}_g(\lambda) := \tilde{r}_g(\lambda) + \sqrt{\frac{2 s_g^2(\lambda) \log(2ML/\delta)}{N_g}} + \frac{7\bar{\rho} \log(2ML/\delta)}{3(N_g - 1)}. \tag{11}$$

Then, with probability at least $1 - \delta$,

$$r_g(\lambda) \le \bar{r}_g(\lambda) \tag{12}$$

for all $g \in \mathcal{G}$ and all $\lambda \in \mathcal{L}$ simultaneously.

Because the bound is uniform over the entire finite table, the controller may query it adaptively. In particular, for any predictable online sequence $\{(g_t, \lambda_t)\}_{t=0}^{T-1}$ taking values in $\mathcal{G} \times \mathcal{L}$,

$$\mathbb{P}\big(e_t \notin \mathcal{E}(g_t, \lambda_t) \mid \mathcal{F}_t\big) \le \bar{r}_{g_t}(\lambda_t) \tag{13}$$

on the same probability-$(1-\delta)$ event. Stage-indexed queries require the same conditional-law assumption for their planning-time residuals; the closed-loop result uses only $\ell = 0$.

Remark 1 (Weights and scope)**.** Sections IV–V require predictable upper miss certificates. Exact ratios or measurable

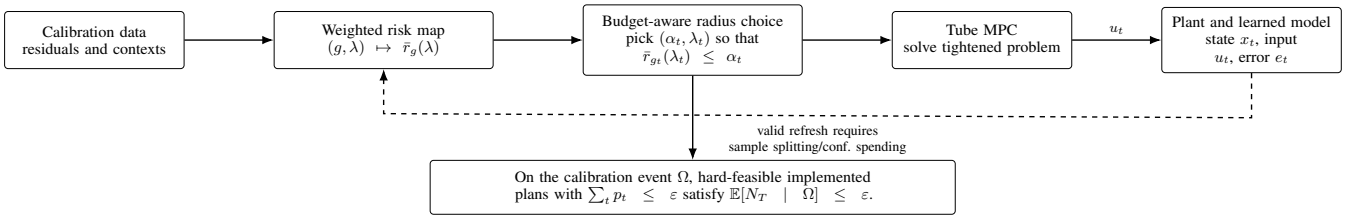


Fig. 1. Closed-loop calibrated robustness as a control architecture. The risk-map layer turns calibration data into a lookup table of local miss-probability upper bounds. The budget manager and radius selector turn those local certificates into tube radii for the controller.

upper weights $\hat{\rho}_g \ge dQ_g^\star/dQ_g^0$, bounded by $\hat{\bar{\rho}}$, are statistical inputs. For upper weights the proof of Theorem 1 uses $\mathbb{E}Z_{g,i} \ge r_g$, so (11) remains valid. Clipping an estimated ratio bounds its range but supplies no upper-envelope guarantee. An independently certified envelope needs its own confidence allocation. Distributional ambiguity sets can instead directly bound miss probabilities; Wasserstein ambiguity sets need not admit density ratios relative to Q_g^0 [3], [21], [22]. Online recalibration requires valid sequential confidence accounting. In this work, we assume that a valid envelope for each feedback-queried conditional law is available.

Rare-miss refinement. Let $k_g(\lambda)$ count the unweighted calibration misses and let $\bar{\rho}_g$ bound the context's weights. Write $b_g^{\mathrm{EB}}(\lambda; d)$ for (11) with confidence error d and cap $\bar{\rho}_g$. The binomial upper limit $U_\tau(k, N)$ is the $(1-\tau)$-quantile of $\mathrm{Beta}(k+1, N-k)$ for $k < N$, and equals 1 for $k = N$ [25]. With $\tau = \delta/(2ML)$, the table

$$\bar{r}_g^\sharp(\lambda_j) = \min_{k \le j}\left\{1,\ b_g^{\mathrm{EB}}(\lambda_k; \delta/2), \bar{\rho}_g U_\tau(k_g(\lambda_k), N_g)\right\} \quad (14)$$

also has simultaneous confidence $1-\delta$. Each bound family receives $\delta/2$. Since miss probabilities are nonincreasing in the radius, any valid upper bound at $\lambda_k \le \lambda_j$ is also valid at λ_j; hence taking the minimum over $k \le j$ preserves validity. The benchmark uses this table in place of $\bar{r}_g$.

IV. Mission-Level Safety Budgeting

Section III gives local miss-probability certificates under the deployment residual law. We now show how they turn into a guarantee on the actual closed loop. At time t, let $M_t := \mathbf{1}\{e_t \notin \mathcal{E}(g_t, \lambda_t)\}$ be the first-step miss event corresponding to the context–radius pair chosen online.

Since the conditional distribution of e_t given $\mathcal{F}_t$ is captured by the deployment residual envelope $Q_{g_t}^\star$, we can apply the bounds derived in Section III. On the good calibration event of Theorem 1,

$$\mathbb{P}(M_t = 1 \mid \mathcal{F}_t) \le p_t := \bar{r}_{g_t}(\lambda_t), \quad (15)$$

where p_t is predictable.

Theorem 2 (Closed-loop cumulative-budget theorem)**.** *Assume Proposition 1 and* (15) *hold on $\Omega \in \mathcal{F}_0$ with $\mathbb{P}(\Omega) > 0$. If the certified local miss probabilities satisfy $\sum_{t=0}^{T-1} p_t \le \varepsilon$ on Ω, then*

$$\mathbb{E}\left[\sum_{t=0}^{T-1} V_t \,\middle|\, \Omega\right] \le \varepsilon. \quad (16)$$

Moreover, for every $\beta \in (0,1)$, with $L_\beta := \log(1/\beta)$,

$$\mathbb{P}\left\{\sum_{t=0}^{T-1} V_t \le \varepsilon + \tfrac{1}{3}L_\beta + \sqrt{2\varepsilon L_\beta + L_\beta^2/9} \,\middle|\, \Omega\right\} \ge 1-\beta. \quad (17)$$

A proof for Theorem 2 is given in Appendix C. Theorem 2 turns safety into bookkeeping: any predictable budget policy is admissible, provided the spent certificates sum to at most ε.

The calibration event is initial information: $\mathbb{P}(\Omega) \ge 1-\delta$ and $\Omega \in \mathcal{F}_0$. Thus the tail statement holds jointly with calibration with probability at least $(1-\delta)(1-\beta)$. For an uncertified continuation, let C_t indicate that certification remains active and the current hard plan is feasible and affordable. This is predictable. Since, on Ω, $\mathbb{P}(C_t V_t = 1 \mid \mathcal{F}_t) \le C_t p_t$, the theorem applies to the stopped violation count $N_T^{\mathrm{hard}} := \sum_t C_t V_t$ with certified risk expenditure $\sum_t C_t p_t$. Hence N_T^{hard} satisfies the same bound, while $N_T \le N_T^{\mathrm{hard}} + \sum_t (1 - C_t)$. For $\varepsilon < 1$, an uncertified fallback has only the trivial conditional violation bound 1. Accounting for such a step would therefore require one full unit of risk budget, which already exceeds the total mission budget.

V. Decision-Aware Radius Selection and the Price of Certification

Theorem 2 specifies how much risk can be spent. In this section we discuss what that spending costs.

For a fixed time t, let $\bar{r}_t(\lambda) := \bar{r}_{g_t}(\lambda)$ and $r_t(\lambda) := r_{g_t}(\lambda)$, with $\Lambda_t \subseteq \Lambda_t^{\mathrm{mpc}}$ the radii considered by the selector.

Given a local target α_t, the controller solves

$$\hat{\lambda}_t \in \mathrm{argmin}_{\lambda \in \Lambda_t} J_t(\lambda) \quad \text{s.t.} \quad \bar{r}_t(\lambda) \le \alpha_t, \quad (18)$$

where $J_t(\lambda)$ is the value of (4) under the common-radius horizon completion of Section II-A; all other planning data are fixed, including in the oracle comparison. The oracle controller, which has perfect knowledge of the true miss curve under the deployment residual law $Q_{g_t}^\star$, solves

$$\lambda_t^\star \in \mathrm{argmin}_{\lambda \in \Lambda_t} J_t(\lambda) \quad \text{s.t.} \quad r_t(\lambda) \le \alpha_t. \quad (19)$$

Ties in (18)–(19) are broken by choosing the smallest feasible radius.

Because larger radii produce tighter constraints, the feasible set of the inner MPC problem is nested in λ, and hence $J_t(\lambda)$ is nondecreasing. To isolate calibration conservatism, the theorem below uses a continuous MPC-feasible radius domain; the finite-grid version additionally requires a feasible upward-rounded comparison radius.

Assumption 1 (Local regularity)**.** *For each t, assume the radius minimizers exist and:*

(a) r_t is nonincreasing and, for some $\eta_t > 0$, the interval $I_t = [\lambda_t^\star, \lambda_t^\star + \eta_t]$ consists of admissible MPC-feasible radii, with $m_t > 0$ such that

$$r_t(\lambda) - r_t(\lambda') \geq m_t(\lambda' - \lambda) \quad \lambda, \lambda' \in I_t, \quad \lambda' \geq \lambda;$$

(b) the error from finite-sample calibration and any upper-weight conservatism satisfies

$$0 \leq \bar{r}_t(\lambda) - r_t(\lambda) \leq \Delta_t \quad \textit{for all feasible } \lambda; \tag{20}$$

(c) the value function is $L_{J,t}$-Lipschitz on the feasible radii:

$$|J_t(\lambda) - J_t(\lambda')| \leq L_{J,t} |\lambda - \lambda'| . \tag{21}$$

Remark 2. These conditions quantify a local sensitivity bound; they require no online estimation. A positive continuous score density on the specified compact interval supplies a positive m_t. The finite displacement used below must remain inside that interval.

Theorem 3 (Decision-aware conservatism theorem)**.** *Under Assumption 1 and $\Delta_t/m_t \leq \eta_t$,*

$$0 \leq \hat{\lambda}_t - \lambda_t^\star \leq \frac{\Delta_t}{m_t}$$

and therefore

$$0 \leq J_t(\hat{\lambda}_t) - J_t(\lambda_t^\star) \leq \frac{L_{J,t}}{m_t}\Delta_t. \tag{22}$$

Summing over the mission yields

$$0 \leq \sum_{t=0}^{T-1} \big(J_t(\hat{\lambda}_t) - J_t(\lambda_t^\star)\big) \leq \sum_{t=0}^{T-1} \frac{L_{J,t}}{m_t}\Delta_t. \tag{23}$$

On a grid, let λ_t^+ be the smallest grid point at or above $\lambda_t^\star + \Delta_t/m_t$. If it exists and is admissible and MPC-feasible, add $h_t := \lambda_t^+ - (\lambda_t^\star + \Delta_t/m_t)$ to the radius bound and $L_{J,t}h_t$ to each cost bound; (20) *and* (21) *must hold at the comparison radii.*

A proof of Theorem 3 is given in Appendix D. The price of certification is composed of three layers: the statistical overestimate Δ_t of the miss curve, the inverse slope m_t^{-1} of the true curve, and the sensitivity $L_{J,t}$ of the inner MPC objective to extra tightening. Better calibration, steeper miss curves, or a less sensitive controller all reduce the gap to the oracle.

VI. A Reproducible Tube-MPC Benchmark

A. Plant, controller, and contexts

We consider a benchmark example which solves (4) for

$$A = \begin{bmatrix} 1 & .15 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} .01125 \\ .15 \end{bmatrix},$$

with $|x^{(1)}| \leq 4$, $|x^{(2)}| \leq 2$, $|u| \leq 1$, and $H = 8$. The stage weights are $Q = \mathrm{diag}(1, .2)$, $R = .08$, with Riccati terminal weight $P = \left[\begin{smallmatrix} 6.36998 & 1.89883 \\ 1.89883 & 1.77191 \end{smallmatrix}\right]$, and $K = [-2.80515 \ -2.68031]$. The context is assigned from $d(x) = \max\{|x^{(1)}|/4, |x^{(2)}|/2\}$ with thresholds $.4$ and $.75$; it is frozen across each planning horizon. The scale $E = .05\xi_g$ in Table I leaves room for input tightening: even at $\lambda = 3.5$, the smallest stage-7 nominal input bound is 0.4622. Each online QP still checks feasibility. Geometries are frozen from independent design samples; deployment uses fresh continuous draws.

B. Methods and metrics

All methods use the horizon-8 MPC formulation with common references and exogenous random draws. Each controller's state determines its visited context. The methods differ in tube geometry and radius selection. SC-MPC uses the context-dependent ellipsoids $\mathcal{E}(g, \lambda)$ with certified risk table $\bar{r}_g^\sharp(\lambda)$ and target $\alpha_t = B_t/(T - t)$. The *reference-risk MPC* uses the same ellipsoids and selection rule with the numerically computed miss probability $r_g(\lambda)$. Matched comparisons at the same state and target isolate certification conservatism; closed-loop costs compare the executed trajectories. The *global-ball* method uses one isotropic set fitted from pooled data. The *fixed-region* method uses the high-context ellipsoid and its empirical miss curve for every state. The *context-ball* method uses context-dependent isotropic sets and allocates more of the remaining budget to later steps, using the formula in Table I. The *fixed-quantile ellipsoid* method selects one radius per context at empirical miss target ε/T, held fixed during the mission. Table I specifies these benchmark variants. Nominal MPC uses zero disturbance tube.

For $n = 500$ missions of $T = 20$ steps, we report the integer-count mean $\widehat{\mathbb{E}N_T} = n^{-1}\sum_j N_T^{(j)}$. Every method uses the common realized cost

$$C = \frac{1}{nT}\sum_{j=1}^{n}\sum_{t=0}^{T-1}\big[\|x_t^{(j)} - x_t^{\mathrm{ref},(j)}\|_Q^2 + R(u_t^{(j)})^2\big].$$

Miss counts refer to disturbance-set escapes on steps with a selected set. Mean set misses use denominator n; fallback frequency uses nT. On loss of affordability or hard feasibility, SC-MPC stops certification and executes nominal-QP recovery, with saturated LQR if that QP fails. All executed steps enter the cost and physical-violation averages.

VII. Numerical Results

A. Operational risk maps and attainability

Figure 2 uses the exact finite table queried by SC-MPC. Reference miss probabilities are computed by radial/angular Student-t quadrature; certification uses the bounded upper weights and is independent of that numerical integration. Increasing the angular resolution changes the reference values by less than 10^{-12}. At the largest radius, the certificates are 5.5854×10^{-5}, 1.7741×10^{-4}, and 0.002442. The exact binomial component reduces the rare-miss margin. For zero calibration misses, it equals $\bar{\rho}_g(1 - \tau^{1/N_g})$, whereas the empirical-Bernstein expression retains its additive finite-sample term. The sample-size study below shows how this statistical margin affects attainability.

TABLE I
HORIZON-8 TUBE-MPC BENCHMARK SPECIFICATION

Radius/data	$\mathcal{L} = \{.4, .6, .8, 1, 1.3, 1.6, 1.9, 2.2, 2.6, 3, 3.5\}$; $\delta = .10$; independent design size 4096 per context; calibration sizes $(N_L, N_M, N_H) = (4, 2, 1)n_0$, $n_0 = 32768$.
Residuals	$E = .05\xi_g$, $\xi_g \sim t_{\nu_g}(0, S_g)$; $\nu = (12, 8, 5)$; $S_L = \left[\begin{smallmatrix} .0025 & .0002 \\ .0002 & .0030 \end{smallmatrix}\right]$, $S_M = \left[\begin{smallmatrix} .0055 & .0006 \\ .0006 & .0045 \end{smallmatrix}\right]$, $S_H = \left[\begin{smallmatrix} .0120 & .0016 \\ .0016 & .0070 \end{smallmatrix}\right]$. These are Student-$t$ scale matrices.
Shift	$w_g(e;\gamma) = 1 + a_g \tanh((\|D_g e\|_2 - 1)/.45)$, $a_g = \min\{.45, b_g\gamma\}$, $b = (.06, .14, .28)$; $D_g = \mathrm{diag}(1/(\alpha_g\sqrt{\mathrm{Var}(E^{(k)})}))$. Fresh rejection draws have density $w_g/\mathbb{E}_0 w_g$ relative to Q_g^0. Calibration uses $\hat{\rho}_g = w_g/(1 - a_g)$ and $\bar{\rho}_g = (1 + a_g)/(1 - a_g)$. Main $\gamma = 1$; shift sweep $.5, 1, 1.5, 2$.
Geometry	With design covariance $\hat{\Sigma}_g$, $M_g^{\mathrm{ell}} = \mathrm{chol}(\alpha_g^2\hat{\Sigma}_g + 10^{-6}I)^{-1}$, $\alpha = (3.6, 2.8, 2.2)$. Context balls use $I/(\beta_g\sqrt{\mathrm{tr}\,\hat{\Sigma}_g/2})$, $\beta = (3, 2.4, 2)$; the global ball uses the pooled design covariance with factor 2.5. All matrices are frozen before calibration.
Compared methods	**SC-MPC**: context-dependent ellipsoids, certified table (14), and equal-share target $\alpha_t = B_t/(T - t)$. **Reference-risk**: same geometry and online radius selection, with the numerical reference miss probability $r_g(\lambda)$ in place of its finite-sample certificate. **Global ball**: one isotropic set fitted to pooled data, with empirical miss target ε/T. **Fixed region**: the high-context ellipsoid and its empirical miss curve are used for every state, with target $\varepsilon/(T - t)$. **Context ball** ("backloaded balls"): context-dependent isotropic sets with the backloaded target specified below. **Fixed-quantile ellipsoid** ("fixed ellipsoids"): one ellipsoid radius per context, fixed at empirical miss target ε/T. **Nominal**: the same MPC problem with zero disturbance tube.
Risk allocation	For the context-ball method, $\alpha_t = B_t(t+1)^{1.6}/\sum_{s=t}^{T-1}(s+1)^{1.6}$. All four empirical comparators use upper-weighted miss rates without finite-sample margins. SC-MPC splits the confidence error equally between the two components of (14). Mission budgets are $.05, .08, .12, .16, .20, .24$.
Initial/reference	$x_0^{(1)} \sim \mathrm{Unif}[-.8, .8]$, $x_0^{(2)} \sim \mathrm{Unif}[-.35, .35]$. References are constant on 0:6, 7:13, 14:19; with probability .75 use random-sign $\mathrm{Unif}[2.2, 3.7]$, otherwise $\mathrm{Unif}[-1.2, 1.2]$; future values are known and the last value is held beyond T.
Solver/seeds	Condensed convex QP via SciPy SLSQP, tolerance 10^{-9}, at most 100 iterations; hard constraints padded by 10^{-6} and independently checked. Seeds: design 101, calibration 2, references 42, initial states 43, deployment $7 + 1000g$ $(g = 0, 1, 2)$. Full tables and per-step ledgers accompany the code.

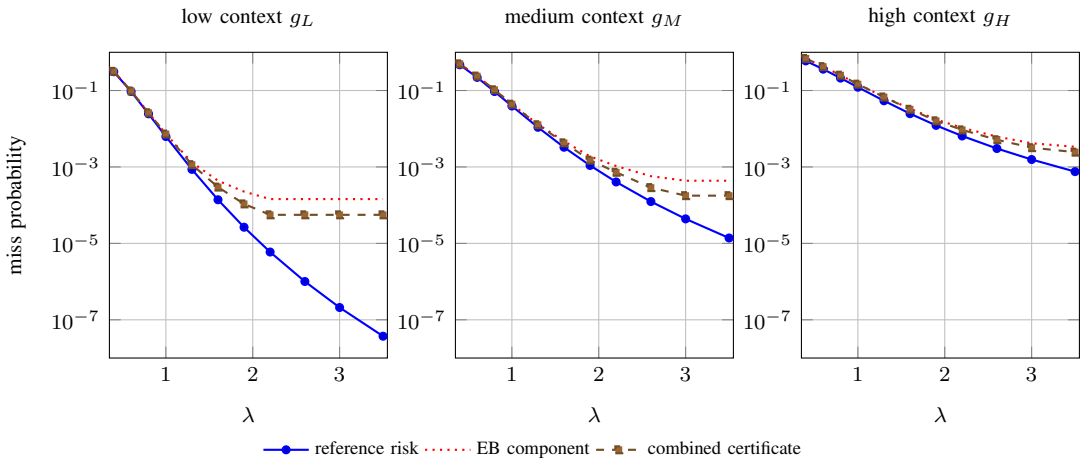


Fig. 2. Operational ellipsoidal risk maps. Solid: numerical reference risk; dotted: empirical-Bernstein component; dashed: combined bound (14) used online. The two components share total confidence error $\delta = .10$.

B. Common cost and matched oracle queries

Figure 3(left) reports the common state/input cost versus the requested mission budget. Table II distinguishes physical violations from tube misses. At $\varepsilon = .12$, the six tube variants recorded zero physical violations, while nominal MPC recorded 12 in 500 missions. Relative safety among the tube variants remains unresolved by these observations. SC-MPC completed all 500 missions at each tested budget with 20 certified steps and no fallback. At $\varepsilon = .12$, the largest total certified risk expenditure $\sum_t p_t$ was 0.119882; at $\varepsilon = .05$, it was 0.049999. These observations describe the sampled missions; Theorem 2 retains its hard-feasibility assumptions.

For the calibration-size study, 240 fixed state/target queries are reused at every sample size. Figure 3(right) shows the fraction admitting both MPC comparisons. At full size, all pairs are feasible and their mean optimal-value gap is 0.047861. The paired records include failed queries. This diagnostic evaluates (22) at a common state and target, while C measures performance along each controller's executed trajectory.

C. Budget records, ablation, and prescribed shift

Figure 4(left) shows an actual certified mission ledger; the straight line is a reference equal-spend schedule. The

TABLE II
BENCHMARK AT $\varepsilon = .12$ (500 MISSIONS)

Method	$\widehat{\mathbb{E}N_T}$	Miss	C	Fallback
Reference risk	0.000	0.138	5.28999	0.0000
SC-MPC	0.000	0.108	5.28996	0.0000
Fixed global ball	0.000	0.050	5.29212	0.0000
Fixed region	0.000	0.032	5.29735	0.0000
Backloaded balls	0.000	0.144	5.29139	0.0000
Fixed ellipsoids	0.000	0.036	5.29028	0.0000
Nominal MPC	0.024	—	5.28254	0.0015

stored records include $(g_t, \lambda_t, p_t, B_t)$ and preserve every budget decrement p_t without clipping. The right panel varies the prescribed within-context reweighting parameter γ. Reference changes alter context visitation; γ separately changes the residual law within each context. The supplied envelope bounds the within-context shift at each γ. With (11) alone, 26 of 500 missions at $\varepsilon = .05$ are not fully certified (none with (14)); the six geometry/selection variants at $\varepsilon = .12$ differ in C by under 0.06%.

VIII. CONCLUSION

This paper proposed a control-oriented framework for closed-loop calibrated robustness in tube MPC. A certified risk map

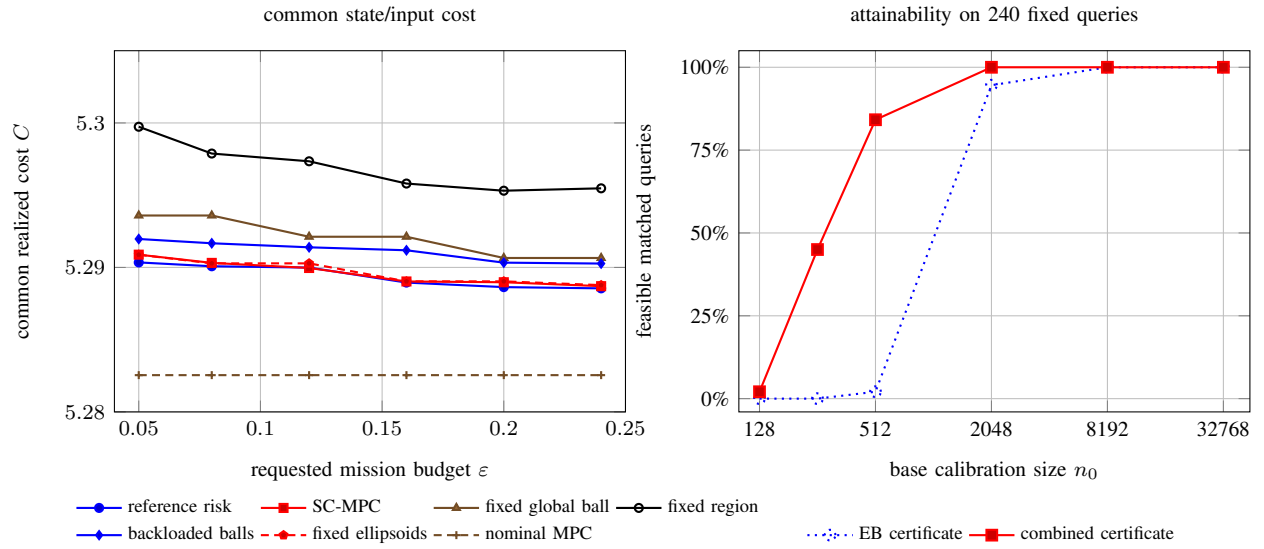


Fig. 3. Left: common realized cost with identical stage weights for every method. Right: attainability of the fixed matched-query set as calibration size increases; infeasible comparisons remain visible.

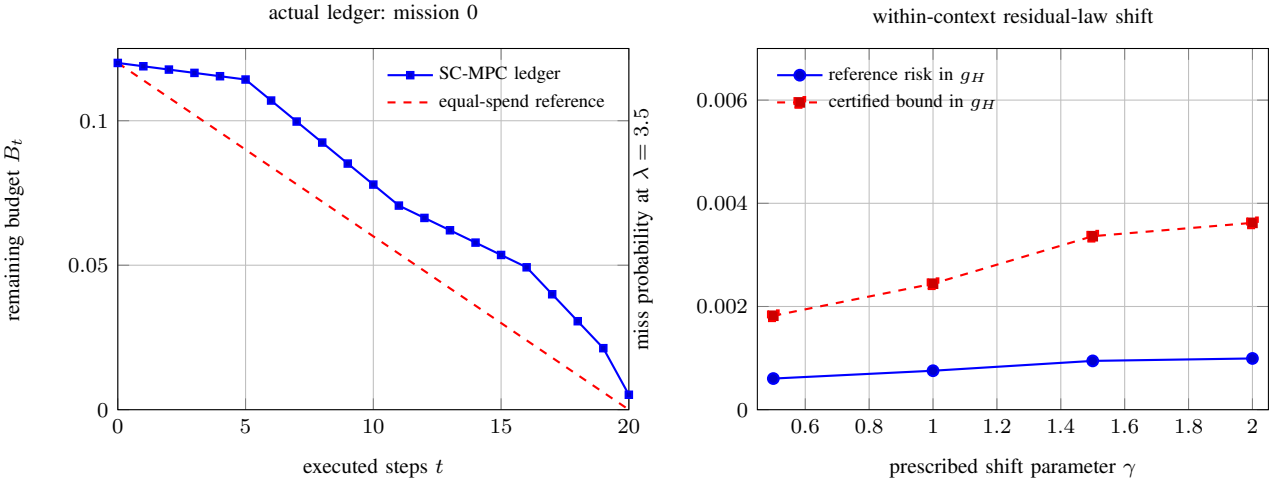


Fig. 4. Left: an executed SC-MPC budget ledger at $\varepsilon = .12$, with an equal-spend reference. Right: high-context reference and certified miss probabilities at $\lambda = 3.5$ under prescribed residual-law shift.

upper-bounds local tube-miss probabilities under a supplied deployment-law envelope. Predictable budget bookkeeping converts these local certificates into mission-level bounds on realized first-step violations, while hard feasibility and the ledger conditions hold. The development focuses on linear or locally linear additive-error models, where the interaction between calibration and robust control can be isolated cleanly. Extension to nonlinear models and online identification of valid policy-dependent envelopes are possible lines of future research.

Appendix A
Proof of Proposition 1

If $e_t \in \mathcal{E}(g_t, \lambda_t)$, then the first tube update in (3) implies $\eta_{t,1} \in \mathcal{T}_{t,1}$. Because the tube-MPC plan is feasible, the tightened constraints ensure that the nominal state–input pair and every error realization in the first tube remain admissible. Hence $u_t \in \mathcal{U}$ and $x_{t+1} = z_{t,1} + \eta_{t,1} \in \mathcal{X}$. The complementary event can therefore occur only if $e_t \notin \mathcal{E}(g_t, \lambda_t)$, which proves (7).

Appendix B
Proof of Theorem 1

Conditioning on the frozen context catalog, grid, matrices M_g, and weights, for fixed g and λ the variables $Z_{g,i}(\lambda)$ defined in (10) are independent, lie in $[0, \bar{\rho}]$, and satisfy

$$\mathbb{E}[Z_{g,i}(\lambda)] = \int \rho_g(e)\mathbf{1}\{\|M_g e\|_2 > \lambda\}\, dQ_g^0(e) = r_g(\lambda).$$

Apply the empirical Bernstein inequality of Maurer and Pontil [23] to the scaled variables $Z_{g,i}(\lambda)/\bar{\rho} \in [0, 1]$. For any fixed (g, λ), with probability at least $1 - \eta$,

$$r_g(\lambda) \le \tilde{r}_g(\lambda) + \sqrt{\frac{2s_g^2(\lambda)\log(2/\eta)}{N_g}} + \frac{7\bar{\rho}\log(2/\eta)}{3(N_g - 1)}.$$

Choosing $\eta = \delta/(ML)$ and taking a union bound over the finite table gives (12). The predictable query statement (13) is immediate because the bound is simultaneous over all $(g, \lambda) \in \mathcal{G} \times \mathcal{L}$. For (14), $k_g(\lambda) \sim \mathrm{Bin}(N_g, q_g)$ with

$q_g = Q_g^0(\|M_g E\| > \lambda)$ and $r_g \le \bar{\rho}_g q_g$. Binomial tail inversion and a union bound allocate $\delta/2$ to these limits. The other $\delta/2$ covers the empirical-Bernstein table; their minimum over both bounds and all $k \le j$ is valid on the intersection.

APPENDIX C
PROOF OF THEOREM 2

Work under $\mathbb{P}(\cdot \mid \Omega)$. Since $\Omega \in \mathcal{F}_0 \subseteq \mathcal{F}_t$, the conditional bound (15) is preserved. Proposition 1 gives $V_t \le M_t$ almost surely. Taking conditional expectations and using (15),

$$\mathbb{E}[V_t \mid \mathcal{F}_t, \Omega] \le \mathbb{E}[M_t \mid \mathcal{F}_t, \Omega] \le p_t. \tag{24}$$

Summing (24) over $t = 0, \dots, T-1$ and taking expectations yields

$$\mathbb{E}\left[\sum_{t=0}^{T-1} V_t \,\middle|\, \Omega\right] \le \mathbb{E}\left[\sum_{t=0}^{T-1} p_t \,\middle|\, \Omega\right] \le \varepsilon,$$

which proves (16).

For the tail bound, let $\xi_t := V_t - \mathbb{E}[V_t \mid \mathcal{F}_t, \Omega]$ and $S_k := \sum_{t=0}^{k-1} \xi_t$. Then S_k is a martingale, $\xi_t \le 1$, and its predictable quadratic variation satisfies

$$W_T := \sum_t \mathrm{Var}(V_t \mid \mathcal{F}_t, \Omega) \le \sum_t \mathbb{E}[V_t \mid \mathcal{F}_t, \Omega] \le \sum_t p_t \le \varepsilon.$$

Freedman's inequality [24] gives $\mathbb{P}\{S_T \ge s \mid \Omega\} \le \exp[-s^2/(2(\varepsilon + s/3))]$. Setting $L_\beta = \log(1/\beta)$ and $s = L_\beta/3 + \sqrt{2\varepsilon L_\beta + L_\beta^2/9}$ makes the exponent at least L_β; combining with $\sum_t V_t \le S_T + \varepsilon$ proves (17).

APPENDIX D
PROOF OF THEOREM 3

Every certified-feasible radius is oracle-feasible, so the smallest minimizers satisfy $\hat{\lambda}_t \ge \lambda_t^\star$. Set $d_t = \Delta_t / m_t$. Since $d_t \le \eta_t$, the comparison point $\lambda_t' = \lambda_t^\star + d_t$ is admissible and MPC-feasible, and $\bar{r}_t(\lambda_t') \le r_t(\lambda_t') + \Delta_t \le r_t(\lambda_t^\star) \le \alpha_t$. Monotonicity of J_t and smallest-radius tie breaking give $\hat{\lambda}_t \le \lambda_t'$. Applying (21) and summing yields (22)–(23). For grid selection, the assumed feasible rounding point obeys $r_t(\lambda_t^+) \le r_t(\lambda_t')$. The error bound then gives $\bar{r}_t(\lambda_t^+) \le \alpha_t$, and $\hat{\lambda}_t - \lambda_t^\star \le d_t + h_t$. The Lipschitz property gives the corresponding cost terms.

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